SOIL EROSION AND CONSERVATION · NOTE ER03

Rainfall Measurement — Rain Gauges, Siting, Network Density and Mean Areal Rainfall

Course topics: SWCE3.5 L4, SWCE3.5 L5, SWCE-505 T6

Measuring rainfall

Rain gauges sample rainfall at selected points. Radar scanning and meteorological-satellite cloud imagery add coverage: radar measures cloud density and rainfall rate, and because drop size correlates with intensity, a radar image gives an approximate rainfall intensity.

Non-recording (standard) rain gauge

- Measures total rainfall (mm) between two readings — usually 24 hours, more often during heavy rain. IMD reads at **08:30 hrs IST**, giving the rainfall of the preceding 24 hours.
- The common Indian type is **Symon's** non-recording gauge. Parts: a circular funnel collector with a metal or brass rim (the notes give 127 mm height and 12.5 mm diameter), funnel of fibreglass-reinforced polyester, collector area 100 cm² or 200 cm², receiving bottle of 2, 4 or 10 L, and a graduated measuring jar.
- Read the bottom of the meniscus at eye level to avoid parallax; clean the funnel, container and glass periodically; mount the gauge on a masonry base.

Check: a 100 cm² or 200 cm² collector is 112.8 mm or 159.6 mm across, so the 12.5 mm figure cannot be the funnel diameter — verify the rim dimensions against IS 5225 before quoting them.

Recording rain gauges

Recording gauges give intensity, frequency, duration and amount — everything that shapes runoff rate and volume. They draw a **mass curve** (cumulative rainfall against time) on a clock-driven chart; its slope is the intensity, and the chart shows when rain started and stopped.

Type	How it works
Float / siphon (most common in India)	IMD specification: catchment diameter 203 mm; float chamber 323.5 ml = 10 mm of rain. A pen on the float stem marks a drum turning once a day or once in 7 days; the chart has numbered hourly lines, 15-minute intermediate lines and horizontal divisions of 0.5 mm rain. At 10 mm the float reaches the top, the siphon empties the chamber (~15 s) and the pen returns to zero — rain during siphoning is lost. Horizontal trace = no rain; gentle slope = low intensity; steep rise = heavy rain; vertical drop = siphoning. Change the chart daily between 08:30 and 09:00 hrs
Tipping bucket	A pair of pivoted buckets under the funnel tips after every 0.25 mm (or 0.1 or 0.2 mm), driving an electronic pen or data logger — suited to digital transmission; used by the US Weather Bureau. Rain smaller than one bucket, rain during tipping and evaporation before tipping are lost; modern units record on magnetic tape or data loggers
Weighing bucket	Records the weight of water collected on a weekly clock-driven chart; the total catch is checked by volume. Not commonly used

Siting, installation and upkeep

- Open, level ground away from buildings and trees — at least twice the obstruction's height away. Avoid steep ridges, narrow valleys and abrupt topography. Funnel opening ≥ 75 cm above ground (as in the notes).
- **Standard gauge:** on a 60 cm × 60 cm masonry foundation set 55 cm below ground; read daily at 08:30 hrs IST, measuring the water in the graduated jar on a level surface; clean regularly and clip vegetation around the site.
- **Recording gauge:** 2–3 m from the standard gauge, perfectly vertical with a horizontal rim 75 cm above ground. Change charts at 08:30–09:00 hrs IST; zero the pen after draining the inner cylinder and add water until siphoning starts; note date and time and number the charts; compare with the nearby standard gauge's daily total; clean, oil and service regularly.

Check: in IMD practice the rim of the standard (non-recording) gauge is 30 cm above ground; the 75 cm height applies to the self-recording gauge.

Rainfall data and network density

- The India Meteorological Department (IMD) is the primary source; data appear in the India Weather Review — Annual Summary, Part A. IMD has also produced 1-hr and 24-hr rainfall maps, normal annual isohyets and maximum-intensity maps for various return periods.
- India has ~10,000 non-recording gauges — about one station per 329 km². WMO recommends 1 per 500 km² in plains and 1 per 150 km² in hills. The network is uneven and much data is unpublished.

For comparison, IS 4987 recommends one gauge per 520 km² in plains, one per 260–390 km² in areas of average elevation up to 1000 m, and one per 130 km² in predominantly hilly areas.

Average rainfall over an area

Method	How	Remarks
Arithmetic mean	Sum of gauge readings ÷ number of gauges	Ignores spatial variation; good only when rainfall is nearly uniform. Reliability depends on gauge spacing, area, rainfall pattern, topography and storm type
Isohyetal	Draw isohyets (lines of equal rainfall) like contours; average = Σ(area between isohyets × mean rainfall of the strip) ÷ total area	Does not fully account for barriers, wind or uneven spacing
Thiessen polygon	Plot gauges on a scaled map, join them with straight lines, draw perpendicular bisectors to form polygons; weight each gauge by its polygon area	More accurate than the arithmetic mean; redraw the polygons whenever gauges change

Thiessen average: $R_ave = (R_1A_1 + R_2A_2 + \ldots + R_5A_5)/(A_1 + A_2 + \ldots + A_5)$

Reading charts

- A recording-gauge chart gives intensity, duration, frequency and the rain/no-rain intervals. Mass curve: horizontal = no rain, steep = intense rain, mild = low intensity.
- The **hyetograph** (intensity against time) is used for short-duration intensity analysis in soil and water conservation.

- Shorter durations have higher intensities; the IDF relation is the most important input for designing conservation works.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. India Meteorological Department; IS 4987, Recommendations for establishing network of raingauge stations. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. IMD reads standard rain gauges daily at

(a) 05:30 hrs (b) 08:30 hrs
(c) 12:00 hrs (d) 17:30 hrs

Q2. The IMD float-type recording gauge siphons after collecting

(a) 2.5 mm (b) 5 mm
(c) 10 mm (d) 25 mm

Q3. Catchment diameter of the IMD siphon gauge is

(a) 127 mm (b) 159 mm
(c) 203 mm (d) 254 mm

Q4. A tipping bucket usually tips after

(a) 0.25 mm (b) 1 mm
(c) 2.5 mm (d) 10 mm

Q5. A rain gauge should be at least ___ the height of an obstruction away from it

(a) equal to (b) twice
(c) four times (d) ten times

Q6. Rain-gauge density in India is about one per

(a) 50 km² (b) 150 km²
(c) 329 km² (d) 1000 km²

Q7. The area-weighted method of averaging rainfall is

(a) arithmetic mean (b) Thiessen polygon
(c) mass curve (d) double mass

Q8. On a siphon-gauge chart, a vertical drop indicates

(a) heavy rain (b) no rain
(c) siphoning (d) clock failure

Answers

Q1 — (b) 08:30 hrs
Q2 — (c) 10 mm
Q3 — (c) 203 mm
Q4 — (a) 0.25 mm
Q5 — (b) twice
Q6 — (c) 329 km²
Q7 — (b) Thiessen polygon
Q8 — (c) siphoning